Period 3 Chlorides (Cambridge (CIE) A Level Chemistry): Revision Note

Reaction of Period 3 Chlorides & Water

- Chlorides of Period 3 elements show characteristic behaviour when added to water which can be explained by looking at their **chemical bonding** and **structure**

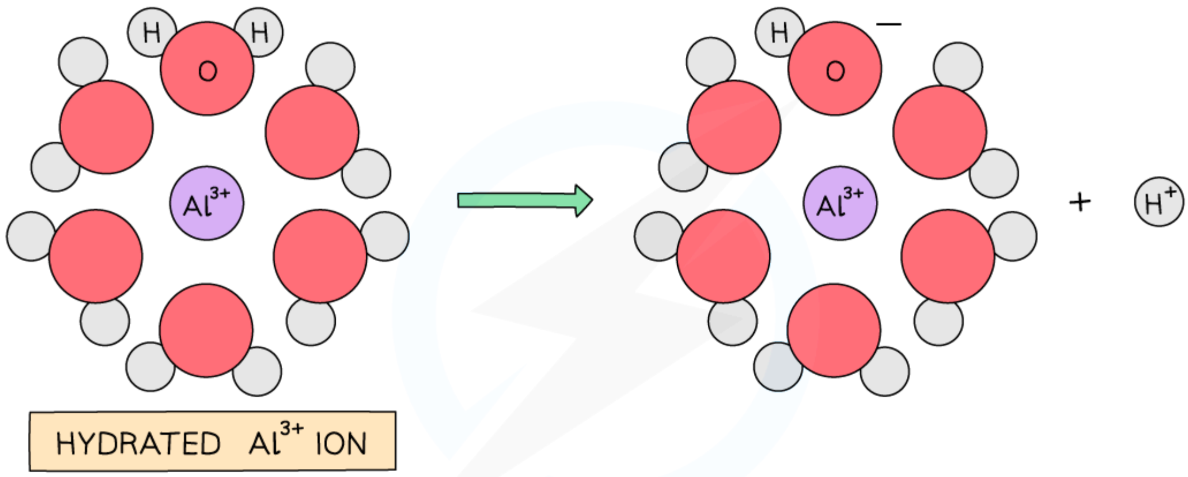
Chemical bonding & structure of Period 3 chlorides table

Period 3 chloride	**$NaCl$**	**$MgCl_2$**	**Al_2Cl_6**	**$SiCl_4$**	**PCl_5**	**SCl_2**
Chemical bonding	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Simple molecular	Simple molecular	Simple molecular	Simple molecular
Observations	White solids dissolve to form colourless solutions		Chlorides react with water giving off white fumes of hydrogen chloride gas			
pH of solution formed	7.0	6.5	3.0	2.0	2.0	2.0

Sodium & magnesium chloride

- $NaCl$ and $MgCl_2$ do not react with water as the **polar** water molecules are attracted to the ions **dissolving** the chlorides and breaking down the **giant ionic structures**: the metal and chloride ions become hydrated ions

How the giant ionic structure of $NaCl$ and $MgCl_2$ break down in water



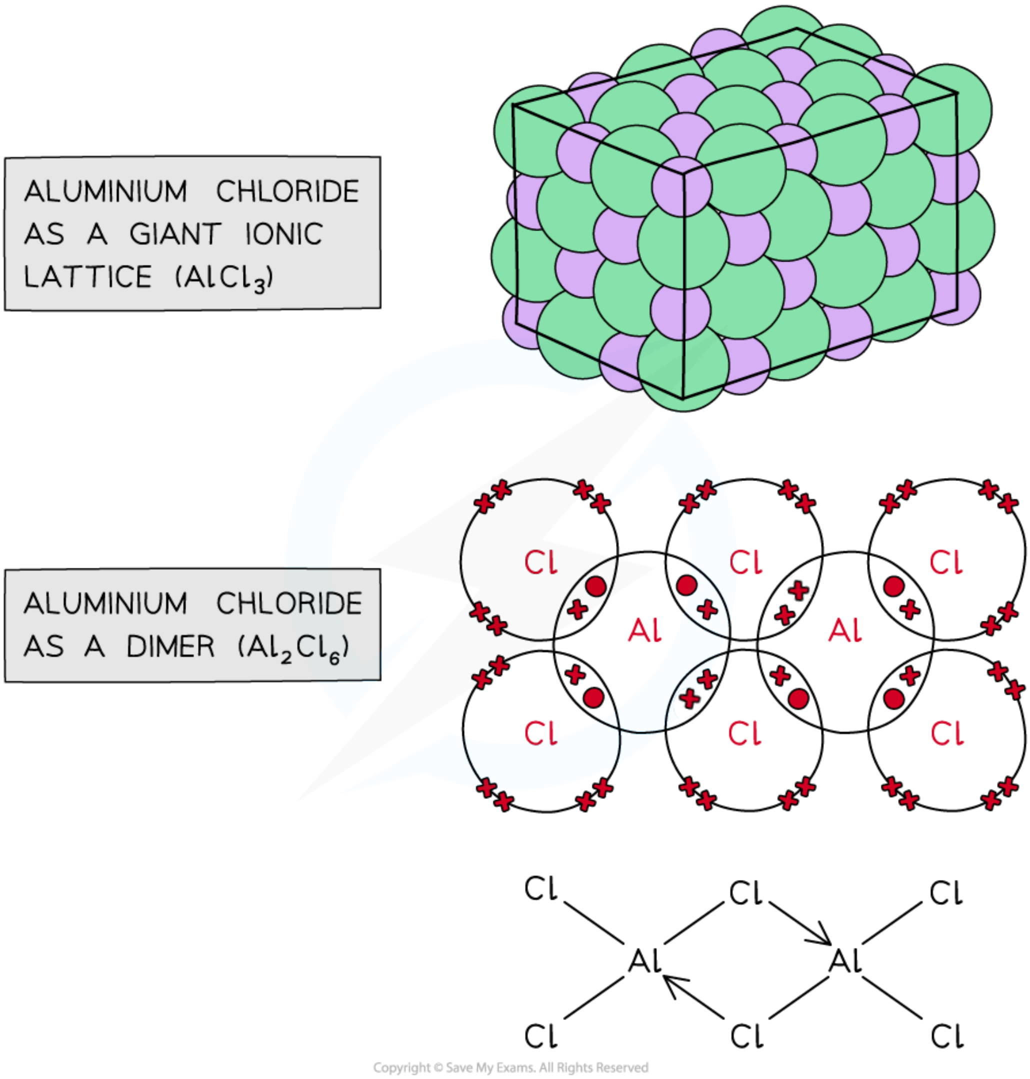
$$[Al(H_2O)_6]^{3+}(aq) \longrightarrow [Al(H_2O)_5OH](aq) + H^+(aq)$$

The hydrated aluminium causes a water molecule to lose a hydrogen ion turning the solution acidic

Aluminium chloride

- Aluminium chloride exists in two forms:
  - $AlCl_3$ is a **giant lattice** with **ionic** bonds
  - Al_2Cl_6 is a **dimer** with **covalent** bonds

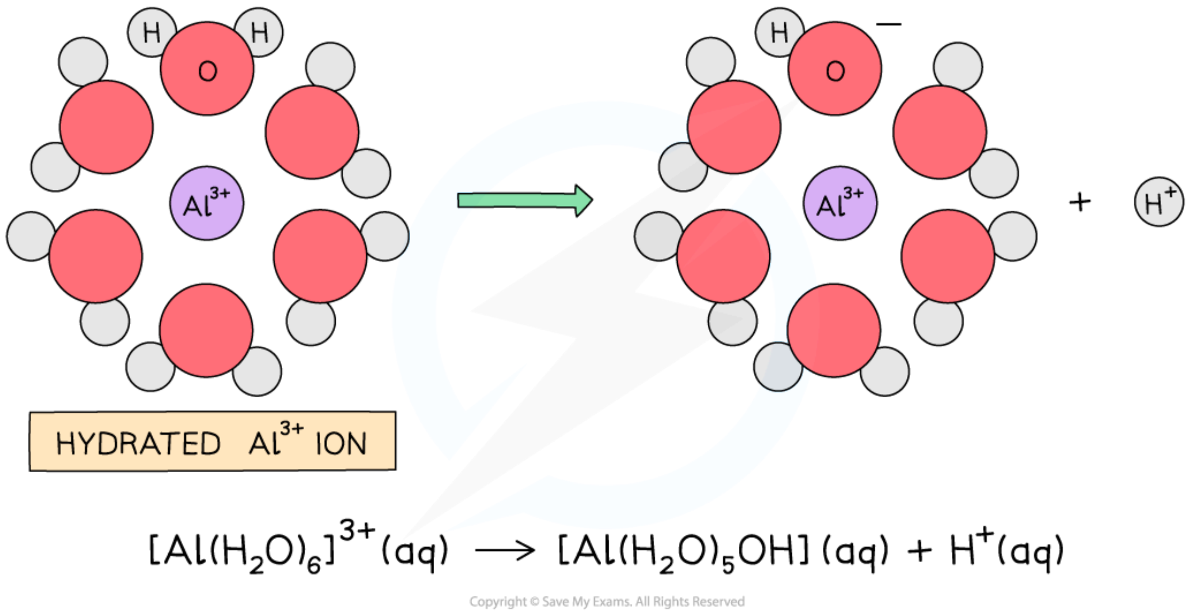
The two forms of aluminium chloride



Aluminium chloride exists as a giant ionic lattice or a covalent dimer

- When water is added to aluminium chloride the dimers are broken down and Al^{3+} and Cl^{-} ions enter the solution
- The highly charged Al^{3+} ion becomes hydrated and causes a water molecule that is bonded to the Al^{3+} to lose an H^{+} ion which turns the solution **acidic**
- The H^{+} and the Cl^{-} form **hydrogen chloride gas** which is given off as **white fumes**

How the Al^{3+} makes an acidic solution



The hydrated aluminium causes a water molecule to lose a hydrogen ion turning the solution acidic

Silicon chloride

- $SiCl_4$ is **hydrolysed** in water, releasing **white fumes of hydrogen chloride gas** in a **rapid** reaction

$$\mathbf{SiCl_4\,(l) + 2H_2O\,(l) \rightarrow SiO_2\,(s) + 4HCl\,(g)}$$

- The SiO_2 is seen as a **white precipitate** and some of the **hydrogen chloride gas** produced dissolves in water to form an **acidic** solution

Phosphorus(V) chloride

- PCl_5 also gets **hydrolysed** in water

$$\mathbf{PCl_5\,(s) + 4H_2O\,(l) \rightarrow H_3PO_4\,(aq) + 5HCl\,(g)}$$

- Both H_3PO_4 and dissolved HCl are highly **acidic**